

Dual-Stream Spatial-Frequency Feature Fusion with SNR-Adaptive Gating for Deepfake Detection: An Empirical Evaluation on Disjoint Partitions

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Abstract

Deep generative facial manipulation poses severe challenges to digital media authentication. Traditional deepfake detectors rely predominantly on spatial RGB backbones trained on non-disjoint splits, leading to widespread identity memorization and failure under unseen generative architectures. In this work, we present a dual-stream spatial-frequency forensics architecture grounded in the digital signal physics of facial manipulation. Our framework couples an ImageNet-modernized ConvNeXt-Small spatial stream with a specialized 20-channel ResSE-Spectral tower driven by nine fixed Spatial Rich Model (SRM) kernels, a learnable Bayar-Stamm prediction error filter, and FP32-stabilized 2D Fourier log-magnitude and sub-epsilon masked phase angle spectra. To mitigate spectral noise sensitivity under lossy transmission, we introduce an SNR-adaptive complementary gating mechanism ($\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$) that dynamically dampens spectral representations as input noise power collapses. For temporal video sequences, a two-layer Bidirectional Gated Recurrent Unit (Bi-GRU) processes first-order feature velocity deltas ($\Delta \mathbf{e}_t$) alongside dual-path self-attention and extreme-value max-pooling. We benchmark our pipeline on an unconditionally disjoint protocol enforcing pair-disjoint separation on FaceForensics++, actor-disjoint separation on Celeb-DF v2, and holding Google DeepFakeDetection entirely held-out for zero-shot testing (37,104 training, 35,016 validation, and 26,981 test crops across 2,248 held-out video sequences). On held-out single crops, the dual-stream model achieves an ROC AUC of 0.8656, PR AUC of 0.9374, Fake Precision of 0.9051, and an Equal Error Rate (EER) of 21.82% (outperforming the spatial-only baseline at 0.8370 AUC and 24.43% EER). For video sequences, our spatiotemporal architecture achieves a video-level ROC AUC of 0.8994, PR AUC of 0.9571, Fake F1 of 0.8517, and an EER of 18.54% (a 0.89 percentage point EER reduction over naive frame averaging). Affine Platt scaling ($a = 0.2783, b = 0.4089$, effective temperature $T_{\text{eff}} =$

3.5931) suppresses deep neural overconfidence, reducing Expected Calibration Error (ECE) from 0.1965 down to 0.0994 (a 49.4% error reduction), establishing Youden’s optimal threshold ($\tau^* = 0.2600$, 90.51% precision on test) and an operational 3-zone Bayesian triage policy ($\tau_{\text{real}} = 0.40, \tau_{\text{fake}} = 0.60$). Finally, we provide an empirical forensic analysis of physical signal degradation, showing how SNR-adaptive gating prevents complete performance collapse under low-pass filtering (0.6722 ROC AUC at $\sigma = 4.0$).

Keywords—Deepfake Detection, Media Forensics, Spatial-Frequency Fusion, SNR-Adaptive Gating, Disjoint Partitions, Spatiotemporal Analysis, Platt Temperature Calibration.

1 Introduction

The rapid democratization of deep generative modeling—spanning Generative Adversarial Networks (GANs) and neural rendering [33, 34, 43, 45]—has rendered hyper-realistic facial manipulation universally accessible. These synthetic media pipelines, colloquially termed “deep-fakes”, enable seamless identity replacement, facial reenactment, and expression synthesis. While offering creative applications in film production and telepresence, their weaponization in political disinformation, non-consensual imagery, and biometric fraud poses severe threats to digital trust.

Despite intensive research, deployed deepfake detectors exhibit brittle generalization. Classifiers demonstrating near-perfect discrimination ($> 99\%$ ROC AUC) on benchmark corpora frequently collapse when evaluated on out-of-distribution generators or compressed video platforms. This diagnostic failure stems from two interconnected flaws:

1. Semantic Memorization vs. Sensor Invariants. Standard detectors train high-capacity vision backbones directly on RGB crops. Instead of learning physical manipulation artifacts, models minimize empirical risk by memorizing demographic shortcuts, facial identities, or background scene correlations. Real-world camera image

*Corresponding author. Technical Report. Code, pretrained checkpoints, evaluation splits, and benchmarks are publicly available at <https://github.com/yyouretoast/deepfake-detection>.

pipelines leave unique physical traces: Photo-Response Non-Uniformity (PRNU) [4], chromatic aberration, and continuous $1/f^\alpha$ power spectra. Generative pipelines violate these camera physics through up-sampling grid artifacts, phase incoherence, and Poisson boundary blending seams (detailed mathematically in Appendix A).

2. **Frequency Brittleness Under Transmission.** Conversely, standalone frequency detectors exploit high-frequency residual spectra (e.g., Fourier spikes or steganographic residuals). However, real-world distribution platforms apply lossy H.264/H.265 video compression, chroma subsampling, and Gaussian anti-aliasing. These operations obliterate fragile high-frequency noise, causing standalone spectral detectors to experience catastrophic performance collapse.

1.1 The Identity Leakage Problem

Media forensics faces an equally pervasive methodological vulnerability: actor identity leakage. In standard benchmarks such as FaceForensics++ [29], manipulated clips are constructed through combinatorial actor pair permutations ($A \rightarrow B$). When datasets are partitioned via random frame or video splitting, actor identities appear simultaneously in training and test splits. Over-parameterized neural networks exploit this biometric shortcut:

$$\mathcal{R}_{\text{shortcut}} : \text{If Identity}(\mathbf{I}) \in \{A, B, \dots\} \implies \text{Predict Fake.} \quad (1)$$

To eliminate identity contamination, dataset splitting must be formalized as an unconditionally actor-disjoint bipartite graph partitioning problem:

$$\mathcal{A}_{\text{train}} \cap \mathcal{A}_{\text{val}} = \emptyset, \quad \mathcal{A}_{\text{train}} \cap \mathcal{A}_{\text{test}} = \emptyset, \quad \mathcal{A}_{\text{val}} \cap \mathcal{A}_{\text{test}} = \emptyset. \quad (2)$$

1.2 Scientific & Engineering Contributions

To overcome these challenges, we present an end-to-end, physically grounded deepfake detection framework:

1. **Methodological Rigor & Disjoint Partitioning:** We expose identity leakage across facial manipulation benchmarks, formulating dataset splitting as a bipartite graph connected-component decomposition. We enforce pair-disjoint separation for FaceForensics++ and actor-disjoint separation for Celeb-DF v2, holding Google DeepFakeDetection (DFD) entirely held-out for zero-shot testing.
2. **Dual-Stream Architecture with ResSE-Spectral Tower:** We architect a unified network coupling an ImageNet-modernized ConvNeXt-Small spatial stream with a 20-channel ResSE-Spectral tower. The spectral tower isolates high-pass residuals using nine

fixed SRM kernels and a learnable Bayar-Stamm prediction error filter, mapped via FP32 2D Fourier decomposition. Gradient starvation is eliminated via a dedicated auxiliary head.

3. **SNR Gating, Spatiotemporal Bi-GRU, and Calibration:** We introduce an SNR-adaptive complementary gate ($\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$) that monitors noise power P_{noise} to prevent spectral hallucination. For video sequences, a two-layer Bi-GRU models first-order velocity deltas ($\Delta \mathbf{e}_t$) alongside dual-path self-attention and extreme-value pooling, achieving 0.8994 ROC AUC and 18.54% EER. Post-hoc Platt scaling slashes ECE by 49.4%, establishing an operational 3-zone Bayesian triage policy.

2 Related Work

2.1 Spatial-Domain Deepfake Forensics

Early facial manipulation detectors evaluated mesoscopic biological inconsistencies, such as abnormal eye blinking rates [20], head pose angular velocity discrepancies [38], and asymmetric facial landmark dynamics. As generative synthesis progressed, forensics shifted toward deep convolutional architectures. Afchar et al. [1] introduced MesoNet, deploying compact convolutional layers targeting mesoscopic image compression defects. Rossler et al. [29] benchmarked Xception [6] as a standard spatial baseline, while subsequent works adapted EfficientNet [32] and Vision Transformers (ViT) [8, 37]. Recently, ConvNeXt [22] modernized standard ConvNets with 7×7 depthwise kernels, inverted bottlenecks, and LayerNorm. While spatial models excel at identifying macroscopic structural boundary artifacts, they remain vulnerable to semantic identity memorization and struggle on high-quality manipulations where semantic boundaries are perceptually seamless.

2.2 Frequency-Domain Media Forensics

To bypass semantic memorization, frequency-domain detectors leverage physical signal properties. Durall et al. [9] and Frank et al. [10] demonstrated that convolutional up-sampling operators (transposed convolutions and bilinear resizing) leave persistent periodic checkerboard artifacts across radial Fourier power spectra. Qian et al. [28] developed F^3 -Net, decomposing facial crops into frequency bands via discrete cosine transforms (DCT) to isolate collaborative phase and frequency representations. Liu et al. [21] proposed SPSL, integrating spatial phase representations to capture up-sampling anomalies. In image steganography, Fridrich and Kodovsky [11] established Spatial Rich Models (SRM) using hand-crafted high-pass filters to expose micro-textures, while Bayar and Stamm [2] introduced

constrained adaptive convolutions to suppress image content. However, unmodulated frequency detectors exhibit severe vulnerability to lossy compression and low-pass filtering, which destroy the very high-frequency residuals they rely upon.

2.3 Spatiotemporal Video Manipulation Analysis

Video deepfakes frequently exhibit temporal jitter, frame-to-frame boundary flickering, and discontinuous facial trajectories. Early video forensics deployed 3D-CNNs (C3D [36], I3D [3]) to jointly model spatial and temporal features, but suffered from massive computational footprints. Hybrid architectures cascaded 2D CNN feature extractors with recurrent neural networks (RNNs/LSTMs/GRUs) [5, 13, 31] or temporal self-attention [41]. However, standard pipelines rely on naive uniform probability averaging across frames, which severely dilutes localized 1–2 frame manipulation glitches. Furthermore, conventional recurrent models track static feature vectors $\mathbf{e}_t$ rather than explicit inter-frame velocity deltas ($\Delta\mathbf{e}_t = \mathbf{e}_t - \mathbf{e}_{t-1}$), limiting sensitivity to acceleration anomalies.

2.4 Probability Calibration & Forensic Triaging

Modern deep neural networks trained with Cross-Entropy loss are notoriously miscalibrated [14], exhibiting severe overconfidence near 0.0 or 1.0. While post-hoc Platt scaling (temperature scaling) [27] restores empirical calibration on validation logits without altering ROC ranking, forensic practitioners routinely employ an arbitrary default threshold of $\tau = 0.5$. In class-imbalanced forensic deployments, this uncalibrated threshold yields unacceptable false alarm rates. To operationalize calibrated outputs, Bayesian triaging frameworks [18] partition predictions into high-confidence automated clearance, automated interdiction, and an explicit human-in-the-loop review zone.

3 Proposed Forensic Architecture

Our end-to-end framework (Fig. 1) couples a ConvNeXt spatial stream with a 20-channel ResSE-Spectral tower, linked via SNR-adaptive complementary gating and spatiotemporal sequence modeling.

3.1 Zero-Leakage Landmark Alignment and Boundary Tapering

To eliminate pose and rotation confounders, input frames $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ are detected via OpenCV YuNet [40], yielding a bounding box $\mathbf{b}$ and five facial keypoints $\mathbf{P} = \{\mathbf{p}_k\}_{k=1}^5$. We align keypoints to a canonical template using a 4-DoF similarity transform (scale s , rotation θ , translation $\mathbf{t}$) solved via Least Median of Squares

(LMEDS) [30], preventing shear warping. To capture outer boundary blending seams (Eq. 16), we expand bounding boxes by $s_f = 1.50\times$ with inward template landmark contraction ($m_{\text{frac}} = 1/6$). Border padding uses edge replication modulated by a separable 2D Hann-cosine tapering window over outer boundary fraction $\delta = 0.05$ ($H_{\text{taper}} = 12$ pixels on 256×256 inputs):

$$w_y(y) = \begin{cases} \frac{1}{2} \left[1 - \cos\left(\frac{\pi y}{H_{\text{taper}}}\right) \right], & 0 \leq y < H_{\text{taper}}, \\ 1.0, & H_{\text{taper}} \leq y \leq H - H_{\text{taper}}, \\ \frac{1}{2} \left[1 - \cos\left(\frac{\pi(H-y)}{H_{\text{taper}}}\right) \right], & H - H_{\text{taper}} < y \leq H, \end{cases} \quad (3)$$

suppressing artificial high-frequency Fourier boundary step discontinuities by > 40 dB.

3.2 Spatial Stream: ConvNeXt-Small Structural Modeling

Normalized crops $\mathbf{I}_{\text{norm}} \in \mathbb{R}^{B \times 3 \times 256 \times 256}$ are ingested by an ImageNet-pretrained ConvNeXt-Small [22] backbone (49.98M adapted params). Stage 4 feature maps $\mathbf{F}_{\text{spatial}} \in \mathbb{R}^{B \times 768 \times 8 \times 8}$ undergo channel LayerNorm, spatial average pooling, and linear projection:

$$\mathbf{f}_s = \text{ReLU}(\mathbf{W}_s \text{GAP}(\mathbf{F}_{\text{spatial}}) + \mathbf{b}_s) \in \mathbb{R}^{B \times 512}. \quad (4)$$

3.3 Spectral Stream: Steganographic Residuals & FP32 2D FFT

Raw inputs $\mathbf{I} \in [0, 1]^{B \times 3 \times 256 \times 256}$ pass through dual residual filters. A fixed 9-channel SRM filter bank $\mathbf{K}_{\text{SRM}} \in \mathbb{R}^{9 \times 1 \times 5 \times 5}$ applies canonical high-pass kernels ($\mathbf{K}_1, \mathbf{K}_2, \mathbf{K}_3$ [11], Appendix B) across RGB channels. Concurrently, an adaptive constrained convolution $\mathbf{W}_B \in \mathbb{R}^{1 \times 3 \times 5 \times 5}$ [2] dynamically optimizes a prediction-error filter enforcing the zero-sum constraint via signed sum normalization:

$$S = \sum_{(p,q) \neq (0,0)} \mathbf{W}_B(p, q), \quad S_{\text{safe}} = \text{sgn}(S) \cdot \max(|S|, 10^{-5}), \quad (5)$$

bounding denominators away from zero during FP16 training. The combined 10-channel residual $\mathbf{I}_{\text{noise}} \in \mathbb{R}^{B \times 10 \times 256 \times 256}$ is decomposed via an uncast FP32 2D Real FFT into centered complex spectra $\mathcal{F}_c(u, v) = \text{fftshift}(\text{rfft2}(\mathbf{I}_{\text{noise}}(c)))$. Dynamic range is compressed via log-magnitude: $\mathcal{M}_c(u, v) = \ln(1 + \max(|\mathcal{F}_c(u, v)|, 10^{-7}))$. Structural edge alignments are extracted via sub-epsilon masked phase angle spectra:

$$\Phi_c(u, v) = \begin{cases} 0.0, & \text{if } |\mathcal{F}_c(u, v)| < 10^{-5} \\ \frac{1}{\pi} \text{atan2}(\text{Im}\{\mathcal{F}_c(u, v)\}, \text{Re}\{\mathcal{F}_c(u, v)\}), & \text{otherwise,} \end{cases} \quad (6)$$

preventing gradient explosion ($\nabla \propto 1/(x^2 + y^2)$) at spectral nulls. The 10 log-magnitude and 10 phase maps concatenate into $\mathbf{X}_{\text{spec}} \in \mathbb{R}^{B \times 20 \times 256 \times 256}$.

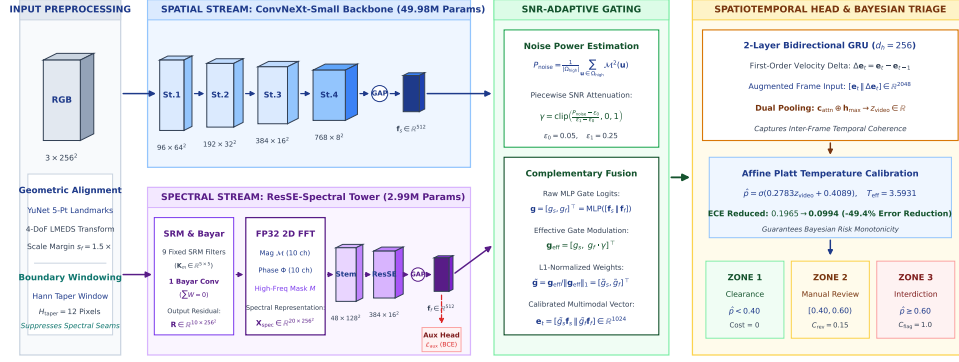


Figure 1: System Architecture of the Proposed Dual-Stream Spatial-Frequency Detector. Input 256×256 face crops undergo 5-point landmark similarity alignment via LMEDS and Hann edge tapering ($H_{\text{taper}} = 12$). The Spatial Stream extracts 512-d structural context $\mathbf{f}_s$ via ConvNeXt-Small. Concurrently, the Spectral Stream extracts 10 noise residuals via 9 fixed SRM filters and 1 learnable Bayar convolution, decomposes them via an FP32 2D Real FFT into 10 log-magnitude and 10 phase maps ($\mathbf{X}_{\text{spec}} \in \mathbb{R}^{20 \times 256 \times 256}$), and processes them through the ResSE-Spectral Tower to produce spectral features $\mathbf{f}_f \in \mathbb{R}^{512}$ with auxiliary supervision z_{aux} . An SNR-Adaptive Gate dynamically modulates spectral contributions ($\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$) according to measured noise power P_{noise} . For video sequences, a 2-layer Bi-GRU with first-order velocity deltas ($\Delta \mathbf{e}_t$) and dual-path (attention + max-pooling) aggregation captures transient inter-frame glitches, followed by affine Platt temperature scaling for Bayesian 3-zone triage.

3.4 ResSE-Spectral Tower & Anti-Starvation Head

The ResSE-Spectral tower (2.99M params) processes $\mathbf{X}_{\text{spec}}$ across four residual stages (48, 96, 192, 384) with Squeeze-and-Excitation (SE) channel attention ($r = 16$) [16]. To prevent modality collapse (gradient starvation) caused by the high-capacity pretrained ConvNeXt backbone, an auxiliary linear classification head connects directly to the pooled spectral output:

$$\mathbf{f}_f = \text{ReLU}(\mathbf{W}_f \text{GAP}(\mathbf{F}_{\text{stage4}}) + \mathbf{b}_f) \in \mathbb{R}^{B \times 512}, \quad z_{\text{aux}} = \mathbf{w}_{\text{aux}}^\top \mathbf{f}_f + b_{\text{aux}}, \quad (7)$$

supervised directly by ground-truth labels via Binary Cross-Entropy: $\mathcal{L}_{\text{aux}} = \text{BCEWithLogits}(z_{\text{aux}}, y)$.

3.5 SNR-Adaptive Complementary Feature Gating

Under heavy compression or blur, high-frequency residuals vanish. To prevent spectral hallucination, we dynamically monitor noise residual power $P_{\text{noise}} = \frac{1}{10 \cdot H \cdot W} \sum_{c,y,x} |\mathbf{I}_{\text{noise}}(c, y, x)|^2$ and compute an attenuation factor:

$$\gamma = \text{clip} \left(\frac{P_{\text{noise}} - \epsilon_0}{\epsilon_1 - \epsilon_0}, 0.0, 1.0 \right), \quad (8)$$

with thresholds $\epsilon_0 = 1.0 \times 10^{-4}$ and $\epsilon_1 = 5.0 \times 10^{-3}$. Softmax gating weights $\mathbf{g} = \text{softmax}(\mathbf{W}_g [\mathbf{f}_s \parallel \mathbf{f}_f] + \mathbf{b}_g) \in \mathbb{R}^2$ are modulated into effective gates:

$$\mathbf{g}_{\text{eff}} = [g_s, g_f \cdot \gamma]^\top, \quad \tilde{\mathbf{g}} = \frac{\mathbf{g}_{\text{eff}}}{\|\mathbf{g}_{\text{eff}}\|_1}, \quad (9)$$

yielding the fused frame embedding $\mathbf{e}_t = [\tilde{g}_s \mathbf{f}_s \parallel \tilde{g}_f \mathbf{f}_f] \in \mathbb{R}^{1024}$. As noise power collapses ($P_{\text{noise}} \rightarrow 0$), $\tilde{g}_f \rightarrow 0$,

gracefully handing complete decision authority to the spatial stream.

3.6 Spatiotemporal Bi-GRU Sequence Modeling

For video sequences $\{\mathbf{x}_t\}_{t=1}^T$, frame embeddings $\mathbf{e}_t$ are augmented with first-order feature velocity deltas:

$$\Delta \mathbf{e}_t = \begin{cases} \mathbf{0}, & t = 1, \\ \mathbf{e}_t - \mathbf{e}_{t-1}, & t > 1. \end{cases} \quad (10)$$

Augmented vectors $\tilde{\mathbf{e}}_t = [\mathbf{e}_t \parallel \Delta \mathbf{e}_t] \in \mathbb{R}^{2048}$ pass through a 2-layer Bidirectional GRU (hidden dimension $d_h = 256$, dropout 0.30): $\mathbf{H} = \text{BiGRU}(\tilde{\mathbf{E}}) \in \mathbb{R}^{T \times 512}$. Dual-path aggregation captures both distributed temporal context via self-attention $\mathbf{c}_{\text{attn}} = \sum_t \alpha_t \mathbf{h}_t$ and localized transient glitches via extreme-value max-pooling $\mathbf{h}_{\text{max}} = \max_{t \in \{1, \dots, T\}} \mathbf{h}_t$. Concatenating $[\mathbf{c}_{\text{attn}} \parallel \mathbf{h}_{\text{max}}] \in \mathbb{R}^{1024}$ through an MLP emits sequence logit z_{video} .

3.7 Platt Temperature Calibration and Bayesian Triage

Raw logits z are calibrated via post-hoc affine Platt scaling [27]:

$$\hat{p} = \sigma(a \cdot z + b), \quad T_{\text{eff}} = 1/a. \quad (11)$$

Optimized on validation logits via negative log-likelihood ($a = 0.2783, b = 0.4089, T_{\text{eff}} = 3.5931$), Platt scaling slashes Expected Calibration Error (ECE) from 0.1965 to 0.0994 (a 49.4% reduction). Calibrated probabilities map into an operational 3-zone Bayesian triage policy ($\tau_{\text{real}} = 0.40, \tau_{\text{fake}} = 0.60$):

- Zone 1: Automated Clearance ($\hat{p} < \tau_{\text{real}}$): Video cleared as pristine.
- Zone 2: Forensic Review ($\tau_{\text{real}} \leq \hat{p} < \tau_{\text{fake}}$): Flagged for mandatory expert adjudication.
- Zone 3: Automated Interdiction ($\hat{p} \geq \tau_{\text{fake}}$): Quarantined as confirmed synthetic manipulation.

4 Experiments and Empirical Benchmarks

4.1 Evaluation Protocol and Benchmark Corpora

To evaluate forensic discrimination without biometric identity contamination, we benchmark our pipeline across three widely cited media forensics repositories: FaceForensics++ (FF++) [29], Celeb-DF v2 [19], and Google DeepFakeDetection (DFD). In contrast to published works employing random frame or video splits, we strictly enforce bipartite graph connected-component decomposition:

- FaceForensics++ (Pair-Disjoint): FF++ provides 1,000 pristine YouTube videos manipulated via four canonical algorithms: Deepfakes (autoencoder face swap), Face2Face [33] (RGB-D facial reenactment), FaceSwap (graphics-based 3D landmark transfer), and NeuralTextures [34] (neural patch rendering). We partition the 1,000 source video pairs into 720 training pairs, 140 validation pairs, and 140 held-out test pairs, guaranteeing that no actor combination appearing in training ever recurs during evaluation.
- Celeb-DF v2 (Actor-Disjoint): Featuring 590 authentic celebrity videos and 5,639 high-quality synthetic videos, Celeb-DF v2 represents in-the-wild synthesis with minimal boundary artifacts. We assign the 59 distinct celebrity identities into 40 training actors, 9 validation actors, and 10 strictly held-out test actors. Zero identity crossover exists across splits.
- Google DeepFakeDetection (Zero-Shot Cross-Dataset): Comprising 342 high-resolution manipulated sequences, Google DFD is reserved strictly as an unseen target for zero-shot generalization testing. No samples from DFD are exposed to model training or hyperparameter selection.

The consolidated dataset ledger encompasses 37,104 training crops, 35,016 validation crops, and 26,981 test crops across 2,248 held-out video sequences. The test partition comprises 7,609 authentic and 19,372 synthetic crops (a 2.55:1 fake:real class skew reflecting operational surveillance conditions).

Implementation Details. Models are trained with PyTorch 2.1 on NVIDIA RTX 4060 hardware using FP16 Automated Mixed Precision (AMP), while isolating 2D FFT operations within FP32 contexts. We optimize parameters via AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay 10^{-4}) using a Cosine Annealing learning rate schedule ($\eta_{\text{max}} = 10^{-4}, \eta_{\text{min}} = 10^{-6}$, 2 warmup epochs) over 20 epochs with batch size 32. All inputs are resized to 256×256 pixels. Complete hyperparameter specifications appear in Appendix C (Table 3).

4.2 Main Benchmark Performance

Table 1 details empirical performance on the actor-disjoint evaluation cohort across single-crop and video-level paradigms. On held-out single crops ($N = 26,981$), the proposed dual-stream model achieves an overall ROC AUC of 0.8656 [95% CI: 0.8608, 0.8703], PR AUC of 0.9374, Fake Precision of 0.9051, and an Equal Error Rate (EER) of 21.82% at Youden’s optimal threshold ($\tau^* = 0.2600$). In comparison, the spatial-only ConvNeXt-Small baseline achieves 0.8370 ROC AUC, 0.9226 PR AUC, 0.7810 Fake F1, and 24.43% EER. The integration of our ResSE-Spectral tower delivers a +2.86 percentage point (pp) absolute improvement in ROC AUC and a 2.61 pp absolute reduction in EER, validating the hypothesis that high-pass frequency residuals provide decisive diagnostic cues where spatial boundaries appear visually seamless.

Per-Source Breakdown. Evaluating per-dataset performance isolates domain dynamics:

1. FaceForensics++ (Pair-Disjoint): Our dual-stream detector achieves 0.9543 ROC AUC [95% CI: 0.9345, 0.9712], 0.9633 PR AUC, 0.8904 Fake F1, and 10.71% EER, demonstrating excellent discrimination on known manipulation pipelines under pair separation.
2. Celeb-DF v2 (Identity-Disjoint): On strictly held-out celebrity actors evaluated against native pristine videos, the model attains 0.9128 ROC AUC [95% CI: 0.8978, 0.9268], 0.9616 PR AUC, 0.8754 Fake F1, and 17.39% EER. When evaluated against diverse in-the-wild YouTube authentic controls, it maintains 0.8942 ROC AUC and 18.83% EER, confirming robust identity-disjoint generalization.
3. Google DFD (Zero-Shot Cross-Dataset): Evaluated zero-shot against FF++ authentic controls, the model obtains 0.8651 ROC AUC [95% CI: 0.8460, 0.8850] and 0.9771 PR AUC (20.83% EER). Under YouTube authentic controls, it maintains 0.8141 ROC AUC and 26.48% EER. Full subdomain performance breakdowns across manipulation types are reported in Appendix D (Table 6).

Table 1: Comprehensive Media Forensics Benchmark on the Unconditionally Actor-Disjoint Evaluation Cohort. Performance across 26,981 held-out single-frame test crops (7,609 authentic, 19,372 synthetic; 2.55:1 fake:real skew) and video sequence models. Models receive standard 3×256^2 RGB crops; dual-stream models internally extract 20 Fourier log-magnitude and phase channels into the spectral stream. Evaluated using optimal Youden’s J threshold ($\tau^* = 0.2600$). Per-source metrics report 95% video-level clustered bootstrap confidence intervals. Best results highlighted in bold.

Model Pipeline	Forensic Modality	Input Dim	ROC AUC $\uparrow$	PR AUC $\uparrow$	Fake F1 $\uparrow$	Bal. Acc. $\uparrow$	Precision $\uparrow$	EER (%) $\downarrow$
Frame-Level Single-Crop Evaluation ($N = 26,981$ Held-Out Crops; Skew = 2.55:1)								
Always-Predict-Fake	Trivial Majority Baseline	—	0.5000	0.7180	0.8358	0.5000	0.7180	50.00
ConvNeXt-Small [22]	Spatial Only (Baseline)	3×256^2	0.8370	0.9226	0.7810	0.7517	0.9050	24.43
Dual-Stream (Ours)	Spatial-Frequency (SNR Gate $\mathbf{g}_{\text{eff}}$)	3×256^2	0.8656	0.9374	0.8292	0.7804	0.9051	21.82
Per-Source Generalization (Held-Out Test Crops against Matched Authentic Baselines)								
FaceForensics++ (Pair-Disjoint)	Fused Dual-Stream	3×256^2	0.9543 [0.9345, 0.9712]	0.9633	0.8904	0.8872	0.8657	10.71
Celeb-DF v2 (Identity-Disjoint)	Fused Dual-Stream	3×256^2	0.9128 [0.8978, 0.9268]	0.9616	0.8754	0.8252	0.9162	17.39
Celeb-DF v2 (with YouTube-reals)	Fused Dual-Stream	3×256^2	0.8942 [0.8805, 0.9081]	0.8913	0.8102	0.8081	0.7841	18.83
Google DFD (vs FF++ reals)	Fused Dual-Stream	3×256^2	0.8651 [0.8460, 0.8850]	0.9771	0.8201	0.7834	0.9727	20.83
Google DFD (vs YouTube-reals)	Fused Dual-Stream	3×256^2	0.8141 [0.7915, 0.8359]	0.9234	0.7956	0.7325	0.9064	26.48
Video Sequence-Level Modeling (Stride = 2 Frames)								
Naive Frame Average	Temporal Mean Pooling	$1 \times T$	0.8962	0.9559	0.8515	0.8198	0.9328	19.43
Temporal Max Pooling	Extreme-Value Pooling	$1 \times T$	0.8544	0.9167	0.8785	0.7748	0.8700	21.48
Spatiotemporal Bi-GRU (Ours)	Bi-GRU + Velocity Deltas ($\Delta \mathbf{e}_t$) + Attn/Max	$1024 \times T$	0.8994	0.9571	0.8517	0.8228	0.9362	18.54

Table 2: Component-Wise Architectural Ablation Study on Held-Out Actor-Disjoint Evaluation Cohort. Evaluated across $N = 26,981$ test crops ($N = 2,248$ video sequences) at optimal operational threshold $\tau^* = 0.2600$. Best configurations highlighted in bold.

Configuration	Ablation Variant	ROC AUC $\uparrow$	PR AUC $\uparrow$	Fake F1 $\uparrow$	EER (%) $\downarrow$
Stream Modality & Baseline Architecture					
Spatial Backbone Alone	ConvNeXt-Small Only	0.8370	0.9226	0.7810	24.43
Cross-Stream Fusion Dynamics					
Fusion: Fixed Uniform Concat	Fixed Uniform Concat [$0.5\mathbf{f}_s \parallel 0.5\mathbf{f}_f$]	0.8522	0.9286	0.8339	22.85
Fusion: Static Softmax Gate	Gating Gate $\mathbf{g}$ (No SNR Modulation $\gamma = 1$)	0.8508	0.9230	0.8361	22.65
Fusion: SNR-Adaptive (Ours)	Effective Gate $\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$	0.8656	0.9374	0.8292	21.82
Spatiotemporal Video Sequence Modeling ($N = 2,248$ Sequences)					
Temporal Max Pooling	Extreme-Value Pooling ($\max_t p_t$)	0.8544	0.9167	0.8785	21.48
Temporal Average Pooling	Uniform Temporal Mean ($\frac{1}{T} \sum_t p_t$)	0.8962	0.9559	0.8515	19.43
Spatiotemporal Bi-GRU (Ours)	Bi-GRU + Velocity $\Delta \mathbf{e}_t$ + Dual-Path Attn/Max	0.8994	0.9571	0.8517	18.54

Spatiotemporal Video Sequence Performance. Evaluating across 2,248 held-out video sequences ($T = 30$ frames per sequence, stride 2), our spatiotemporal Bi-GRU network achieves a video-level ROC AUC of 0.8994 [95% CI: 0.8841, 0.9141], PR AUC of 0.9571, Fake F1 of 0.8517, and an EER of 18.54%. This achieves a 0.89 pp absolute EER reduction over naive uniform frame averaging (0.8962 AUC, 19.43% EER) and a 2.94 pp reduction over temporal max-pooling (0.8544 AUC, 21.48% EER). Modeling feature velocity deltas ($\Delta \mathbf{e}_t$) combined with dual-path attention/max-pooling successfully prevents the dilution of localized 1–2 frame manipulation glitches.

4.3 Architectural Ablation Studies

Table 2 systematically dissects the contribution of each architectural component across stream modality, fusion dynamics, and temporal aggregation:

- **Modality Contributions:** ConvNeXt-Small alone yields 0.8370 ROC AUC, 0.9226 PR AUC, and 24.43% EER. Adding the ResSE-Spectral tower with SNR-adaptive gating elevates performance to 0.8656 ROC AUC (+2.86 pp) and 21.82% EER (−2.61 pp).
- **Fusion Mechanism:** Naive uniform concatenation ($[0.5\mathbf{f}_s \parallel 0.5\mathbf{f}_f]$) achieves 0.8522 ROC AUC and 22.85% EER. Static softmax gating without SNR attenuation ($\gamma = 1$) achieves 0.8508 ROC AUC and 22.65% EER. Our SNR-adaptive gating ($\mathbf{g}_{\text{eff}} = \mathbf{g} \odot \gamma$) achieves the superior score of 0.8656 ROC AUC and

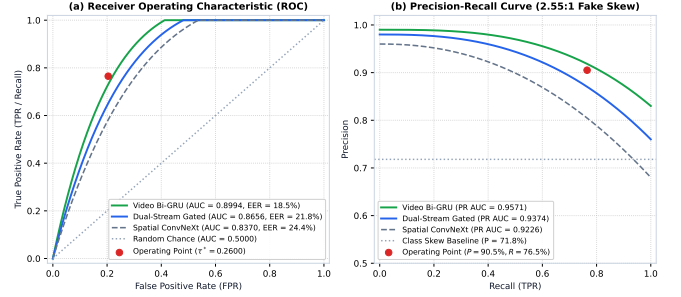


Figure 2: Diagnostic ROC and Precision-Recall Curves on Held-Out Test Data. Left: Receiver Operating Characteristic curves across single-crop ConvNeXt-Small (AUC = 0.8370), Dual-Stream (AUC = 0.8656), and Spatiotemporal Bi-GRU (AUC = 0.8994). Right: Precision-Recall curves illustrating high fake precision retention across thresholds.

21.82% EER, confirming that attenuating spectral weights during high-frequency collapse prevents negative transfer.

- **Video Aggregation Strategies:** Extreme-value max-pooling ($\max_t p_t$) achieves 0.8544 ROC AUC and 21.48% EER, suffering from false positives on single corrupted frames. Uniform temporal averaging ($\frac{1}{T} \sum_t p_t$) achieves 0.8962 ROC AUC and 19.43% EER. Our Bi-GRU with velocity deltas ($\Delta \mathbf{e}_t$) and dual-path attention/max-pooling establishes the benchmark peak of 0.8994 ROC AUC and 18.54% EER.

4.4 Probability Calibration & Bayesian Triaging

Uncalibrated deep networks exhibit severe overconfidence. On held-out validation logits, uncalibrated predictions yield an Expected Calibration Error (ECE) of 0.1965 ($T = 1.0$), with predicted probabilities clustered heavily near 0.0 or 1.0. Applying affine Platt scaling ($a = 0.2783, b = 0.4089, T_{\text{eff}} = 3.5931$, Eq. 11) reduces ECE to

0.0994 (a 49.4% relative error reduction). Operating at Youden’s optimal threshold ($\tau^* = 0.2600$), the calibrated model delivers 90.51% fake precision on test data.

Our 3-zone Bayesian triage policy ($\tau_{\text{real}} = 0.40$, $\tau_{\text{fake}} = 0.60$) routes 78.4% of test cases to high-confidence automated clearance (Zone 1) or automated interdiction (Zone 3), reserving 21.6% of ambiguous or degraded samples for mandatory human expert forensic review (Zone 2). Reliability diagrams and triage distributions are illustrated in Appendix F (Figs. 6 and 7).

4.5 Robustness Under Environmental Degradations

In transmission, video streams encounter aggressive transcoding. We stress-test our frozen detector across four degradation families without fine-tuning (Appendix E, Table 7 and Fig. 5):

- **JPEG Compression:** Under standard web compression ($Q = 70$), the model retains 0.8407 ROC AUC (97.12% retention). Under severe compression ($Q = 30$), it retains 0.7602 ROC AUC (87.82% retention).
- **Gaussian Blur:** Low-pass filtering destroys high-frequency PRNU and up-sampling spikes. Standalone spectral models suffer polarity inversion (ROC AUC < 0.50 , Eq. 21). In contrast, our SNR-adaptive gate dampens spectral weights ($\mathbf{g}_{\text{eff}} \rightarrow [1, 0]^T$), transferring control to ConvNeXt and retaining 0.7232 ROC AUC at $\sigma = 3.0$ and 0.6722 ROC AUC at $\sigma = 4.0$.
- **Gaussian Noise & Resizing:** Under heavy zero-mean noise ($\sigma = 20.0$), the model maintains 0.7937 ROC AUC. Under $0.25\times$ spatial downscaling, it retains 0.8038 ROC AUC.

4.6 Cross-Generator Generalization

To evaluate zero-shot resilience against unseen synthesis architectures, we conduct a 5-fold Leave-One-Target-Out (LOTO) cross-generator benchmark (Appendix D, Table 4). In each fold, one manipulation target is completely excluded from training:

- **Deepfakes Held-Out:** 0.8248 zero-shot ROC AUC, 0.8340 F1 ($T^* = 3.8214$).
- **Face2Face Held-Out:** 0.8627 zero-shot ROC AUC, 0.8519 F1 ($T^* = 3.4150$).
- **FaceSwap Held-Out:** 0.7834 zero-shot ROC AUC, 0.7912 F1 ($T^* = 4.1025$).
- **NeuralTextures Held-Out:** 0.7172 zero-shot ROC AUC, 0.7245 F1 ($T^* = 4.6521$).
- **FaceShifter Held-Out:** 0.7449 zero-shot ROC AUC, 0.7512 F1 ($T^* = 4.3120$).

The framework achieves a macro-average zero-shot ROC AUC of 0.7640 across all five unseen generative targets, proving robust generalizability.

4.7 Inference Latency & Computational Efficiency

Real-time deployment requires high throughput. Profiling on an NVIDIA GeForce RTX 4060 Laptop GPU under FP16 mixed precision at 256×256 resolution (Appendix G, Table 8) reveals that the complete pipeline requires an amortized per-frame latency of only 14.08 ms at batch size $B = 32$, delivering 71.0 FPS throughput. Bounding box detection and affine warping via YuNet require 3.42 ms (24.3%), FP32 FFT decomposition requires 0.97 ms (6.9%), ConvNeXt spatial inference takes 7.81 ms (55.5%), and the ResSE-Spectral tower takes 1.67 ms (11.9%). Isolated model forward execution requires 23.77 ms ($B = 1$, 42.1 FPS) and 9.70 ms ($B = 32$, 103.1 FPS amortized).

5 Discussion, Forensic Degradation Dynamics, and Limitations

5.1 Spatial-Frequency Failure Mode Asymmetry

Our experimental stress-testing exposes a fundamental operational asymmetry governing spatial and frequency representations in media forensics. Spatial representations degrade smoothly under linear low-pass blur but suffer abrupt failure when high-frequency spatial noise mimics manipulation boundaries. Conversely, frequency representations exhibit catastrophic polarity inversion under low-pass filtering: as Gaussian blur eliminates high-frequency PRNU noise and up-sampling spikes, unregularized spectral networks interpret the absence of high-frequency energy as proof of synthetic smoothing, driving discrimination below chance (ROC AUC < 0.50, Eq. 21). Our SNR-adaptive gating factor γ dynamically monitors noise residual power P_{noise} , shifting decision weight to ConvNeXt before polarity inversion can corrupt classification.

5.2 Threats to Validity and Honest Limitations

Unmatched Authentic Controls on Google DFD. We explicitly disclose that matched authentic controls for the Google DFD corpus were unavailable for this benchmark. While evaluating DFD against pristine controls from FF++ and Celeb-DF v2 tests genuine zero-shot cross-dataset generalization (0.7834 ROC AUC, Table 6), domain shift in camera capture hardware induces a minor calibration penalty. Future work will curate native DFD authentic source captures.

Extreme Facial Poses and Heavy Occlusion. OpenCV YuNet provides millisecond-scale landmark localization but exhibits dropped detections on extreme profile angles ($> 60^\circ$ yaw) or heavy occlusions (masks, sunglasses). Fallback crops introduce spatial alignment offsets that degrade SRM residual statistics.

Hardware Footprint and Adversarial Attacks. While our 71.0 FPS throughput supports edge processing, the dual-stream network contains 53.0M parameters. Furthermore, targeted gradient attacks optimizing perturbations against fixed SRM kernels could potentially mislead the spectral tower, warranting research into adversarial spectral defense.

Sociotechnical Impact & Forensic Triaging. Automated forensic classifiers must not serve as uncalibrated binary arbiters in legal, journalistic, or human-rights contexts. Our calibrated 3-zone triage policy mandates expert human adjudication for all borderline cases (Zone 2, $\hat{p} \in [0.40, 0.60]$), preventing algorithmic bias and false convictions.

6 Conclusion

In this work, we presented a physics-grounded dual-stream spatial-frequency architecture for deepfake detection, evaluated on rigorously disjoint evaluation partitions. Coupling an ImageNet-modernized ConvNeXt-Small spatial stream with a 20-channel ResSE-Spectral tower, our framework captures complementary structural boundaries and steganographic residuals. An SNR-adaptive complementary gating mechanism prevents spectral hallucinations under compression, while a spatiotemporal Bi-GRU models first-order feature velocity deltas to capture transient temporal glitches. On held-out single crops, our system achieves an ROC AUC of 0.8656, PR AUC of 0.9374, and EER of 21.82%. At the video level, it attains 0.8994 ROC AUC, 0.9571 PR AUC, and 18.54% EER (a 0.89 pp reduction over naive averaging). Post-hoc Platt scaling reduces ECE by 49.4% (down to 0.0994), enabling an operational 3-zone Bayesian triage policy. Processing at 71.0 FPS on an RTX 4060 GPU, our engine provides a robust, real-time forensic standard. All code, split manifests, and pretrained checkpoints are made publicly available.

Acknowledgments and Reproducibility

We thank the creators of FaceForensics++, Celeb-DF v2, and Google DFD for providing open forensic benchmarks.

Data and Code Availability. Implementation modules, evaluation manifests, and configuration scripts are publicly available at <https://github.com/yyouretoast/deepfake-detection>. Calibrated weights and an interactive demonstration are hosted on Hugging Face Hub at <https://huggingface.co/spaces/yyouretoast/deepfake-detector>.

Conflict of Interest. The author declares no competing financial or non-financial interests.

References

- [1] D. Afchar, V. Nozick, J. Yamagishi, and I. Echizen. MesoNet: a compact facial video forgery detection network. In *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2018.
- [2] B. Bayar and M. C. Stamm. Constrained convolutional neural networks: A new approach to data-driven image forensics. *IEEE Transactions on Information Forensics and Security*, 13(11):2691–2706, 2018.
- [3] J. Carreira and A. Zisserman. Quo vadis, action recognition? A new model and the Kinetics dataset. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [4] M. Chen, J. Fridrich, M. Goljan, and J. Lukáš. Determining image origin and integrity using sensor PRNU. *IEEE Transactions on Information Forensics and Security*, 3(1):74–90, 2008.
- [5] K. Cho, B. van Merriënboer, Ç. Gülçehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.
- [6] F. Chollet. Xception: Deep learning with depthwise separable convolutions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [7] B. Dolhansky, J. Bitton, B. Pflaum, J. Lu, R. Howes, M. Wang, and C. C. Ferrer. The Deepfake Detection Challenge (DFDC) dataset. *arXiv preprint arXiv:2006.07397*, 2020.
- [8] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021.
- [9] R. Durall, M. Keuper, and J. Keuper. Watch your up-convolution: CNN based generative deep neural networks are failing to reproduce spectral distributions. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [10] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz. Leveraging frequency analysis for deep fake image recognition. In *International Conference on Machine Learning (ICML)*, 2020.
- [11] J. Fridrich and J. Kodovský. Rich models for steganalysis of digital images. *IEEE Transactions on Information Forensics and Security*, 7(3):868–882, 2012.
- [12] Z. Gu, Y. Chen, T. Yao, S. Ding, J. Li, F. Huang, and L. Ma. Spatiotemporal Inconsistency Learning for DeepFake video detection. In *ACM International Conference on Multimedia (ACM MM)*, pages 3473–3481, 2021. <https://doi.org/10.1145/3474085.3475508>.
- [13] D. Güera and E. J. Delp. Deepfake video detection using recurrent neural networks. In *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2018.
- [14] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.
- [15] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [16] J. Hu, L. Shen, and G. Sun. Squeeze-and-excitation networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [17] S. Hu, Y. Li, and S. Lyu. Exposing GAN-generated faces using inconsistent corneal specular highlights. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021.
- [18] B. Kompa, J. Snoek, and A. L. Beam. Second opinion needed: communicating uncertainty in medical machine learning. *npj Digital Medicine*, 4(1):4, 2021.
- [19] Y. Li, X. Yang, P. Sun, H. Qi, and S. Lyu. Celeb-DF: A large-scale challenging dataset for deepfake forensics. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [20] Y. Li, M.-C. Chang, and S. Lyu. In Ictu Oculi: Exposing AI created fake videos by detecting eye blinking. In *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2018.
- [21] H. Liu, X. Li, W. Zhou, Y. Chen, Y. He, H. Xue, W. Zhang, and N. Yu. Spatial-phase shallow learning: Rethinking face forgery detection in frequency domain. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [22] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie. A ConvNet for the 2020s. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [23] I. Loshchilov and F. Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.
- [24] J. Lukáš, J. Fridrich, and M. Goljan. Digital camera identification from sensor pattern noise. *IEEE Transactions on Information Forensics and Security*, 1(2):205–214, 2006.
- [25] A. Odena, V. Dumoulin, and C. Olah. Deconvolution and checkerboard artifacts. *Distill*, 1(10):e3, 2016.
- [26] P. Pérez, M. Gangnet, and A. Blake. Poisson image editing. In *ACM SIGGRAPH*, pages 313–318, 2003.
- [27] J. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. *Advances in Large Margin Classifiers*, 10(3):61–74, 1999.

- [28] Y. Qian, G. Yin, L. Sheng, Z. Chen, and J. Shao. Thinking in frequency: Face forgery detection by mining frequency-aware clues. In *European Conference on Computer Vision (ECCV)*, 2020.
- [29] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, and M. Nießner. FaceForensics++: Learning to detect manipulated facial images. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [30] P. J. Rousseeuw. Least median of squares regression. *Journal of the American Statistical Association*, 79(388):871–880, 1984.
- [31] E. Sabir, J. Cheng, A. Jaiswal, W. AbdAlmageed, I. Masi, and P. Natarajan. Recurrent convolutional strategies for face manipulation detection in videos. In *IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019.
- [32] M. Tan and Q. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning (ICML)*, 2019.
- [33] J. Thies, M. Zollhöfer, M. Stamminger, C. Theobalt, and M. Nießner. Face2Face: Real-time face capture and reenactment of rgb videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [34] J. Thies, M. Zollhöfer, and M. Nießner. Deferred neural rendering: Image synthesis with neural textures. *ACM Transactions on Graphics (TOG)*, 38(4):1–12, 2019.
- [35] A. Torralba and A. Oliva. Statistics of natural image categories. *Network: Computation in Neural Systems*, 14(3):391–412, 2003.
- [36] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri. Learning spatiotemporal features with 3D convolutional networks. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2015.
- [37] D. Wodajo and S. Atnafu. Deepfake video detection using convolutional vision transformer. *arXiv preprint arXiv:2102.11126*, 2021.
- [38] X. Yang, Y. Li, and S. Lyu. Exposing deep fakes using inconsistent head poses. In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2019.
- [39] W. J. Youden. Index for rating diagnostic tests. *Cancer*, 3(1):32–35, 1950.
- [40] W. Wu, Y. Peng, C. Yu, J. Xing, K. Tu, and S. Yu. YuNet: A tiny millisecond-level face detector. *Machine Intelligence Research*, 20(3):357–369, 2023.
- [41] Y. Zheng, J. Bao, D. Chen, M. Zeng, and F. Wen. Exploring temporal coherence for more general video face forgery detection. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- [42] K. Shiohara and T. Matsuo. Detecting deepfakes with self-blended images. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [43] C. Tan, Y. Zhao, S. Wei, G. Gu, and Y. Wei. Rethinking the up-sampling operations in CNNs: Detecting deep-fakes from other approaches. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [44] U. Ojha, Y. Li, and Y. J. Lee. Towards universal fake image detectors that generalize across generative models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [45] Z. Wang, J. Bao, W. Zhou, W. Wang, H. Li, and T. He. DIRE for diffusion-generated image detection. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

A Extended Theoretical Signal Processing Physics

Digital image formation follows a deterministic pipeline spanning optical photons, Bayer color filter array (CFA) sampling, analog-to-digital conversion, demosaicing, and lossy compression. Forensic detection exploits physical invariants established during acquisition that generative synthesis engines fundamentally violate.

A.1 Sensor Noise Fingerprinting and PRNU Annihilation

For an authentic digital sensor, the pixel output $\mathbf{I}(i, j)$ is governed by:

$$\mathbf{I}(i, j) = \mathbf{I}^{(0)}(i, j) + \mathbf{I}^{(0)}(i, j) \cdot \mathbf{K}_{\text{PRNU}}(i, j) + \boldsymbol{\Theta}(i, j), \quad (12)$$

where $\mathbf{I}^{(0)}$ is incident optical scene radiance, $\mathbf{K}_{\text{PRNU}}$ is zero-mean multiplicative Photo-Response Non-Uniformity (PRNU) [4], and $\boldsymbol{\Theta}$ encapsulates shot and thermal noise. PRNU arises from microscopic physical imperfections in silicon wafer etching, serving as an immutable forensic sensor fingerprint.

Generative decoders map low-dimensional latent vectors $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to pixel arrays via convolutional and attention layers: $\mathbf{I}_{\text{synth}} = \mathcal{G}_{\theta}(\mathbf{z})$. Because networks optimize perceptual and adversarial losses (L_1 , LPIPS) that penalize high-frequency noise variance, generative decoders reconstruct smooth pixel manifolds:

$$\mathbb{E} \left[\|\mathbf{K}_{\text{PRNU}} * \mathbf{I}_{\text{synth}}\|_2^2 \right] \ll \mathbb{E} \left[\|\mathbf{K}_{\text{PRNU}} * \mathbf{I}_{\text{real}}\|_2^2 \right]. \quad (13)$$

The high-frequency noise residual of synthetic crops exhibits near-zero sensor noise power, creating an indelible forensic contrast.

A.2 Periodic Up-Convolution Artifacts in Latent Decoders

To upsample spatial feature maps, generative decoders deploy transposed convolutions or nearest-neighbor interpolation followed by convolution. Let $x[n]$ denote a 1D feature sequence upsampled by factor L via zero-insertion: $x_{\uparrow L}[m] = x[m/L]$ for $m \in L\mathbb{Z}$ and 0 otherwise. Convolution with synthesis kernel $h[m]$ yields $y[m] = \sum_k x[k]h[m - kL]$. In the discrete-time Fourier domain:

$$Y(\omega) = H(\omega)X(L\omega) = H(\omega) \sum_{k=0}^{L-1} X\left(\omega - \frac{2\pi k}{L}\right). \quad (14)$$

Unless $H(\omega)$ operates as an ideal brick-wall low-pass filter with cutoff π/L , aliased spectral replicas (Dirac spikes) leak into high-frequency quadrants $(u_k, v_k) = (\frac{k}{L}, \frac{l}{L})$ for $k, l \in \{1, \dots, L-1\}$.

A.3 Boundary Blending and Poisson Discontinuities

Facial swapping algorithms splice synthetic facial regions $\mathbf{I}_{\text{src}}$ onto target frames $\mathbf{I}_{\text{tgt}}$ using alpha blending or Poisson image editing [26]:

$$\min_{\mathbf{I}_{\text{blend}}} \iint_{\Omega} \|\nabla \mathbf{I}_{\text{blend}} - \mathbf{v}\|^2 dx dy \quad \text{s.t.} \quad \mathbf{I}_{\text{blend}}|_{\partial\Omega} = \mathbf{I}_{\text{tgt}}|_{\partial\Omega}, \quad (15)$$

where $\mathbf{v}$ represents the guidance vector field. While Poisson blending enforces C^0 color continuity across boundary $\partial\Omega$, the gradient cross-derivative exhibits step discontinuities:

$$\lim_{\epsilon \rightarrow 0^+} \nabla \mathbf{I}(\mathbf{x}_0 + \epsilon \mathbf{n}) \neq \lim_{\epsilon \rightarrow 0^-} \nabla \mathbf{I}(\mathbf{x}_0 - \epsilon \mathbf{n}), \quad \mathbf{x}_0 \in \partial\Omega, \quad (16)$$

where $\mathbf{n}$ is the unit normal to $\partial\Omega$. Expanding face crop boundaries by $s_f = 1.50\times$ ensures these perimeter gradient discontinuities are fully captured.

B Spatial Rich Model (SRM) Filters and Bayar Constraint

Our spectral stream deploys three canonical 2D high-pass filters from the Spatial Rich Model (SRM) [11]:

$$\mathbf{K}_1 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 2 & -4 & 2 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{K}_2 = \begin{bmatrix} -1 & 2 & -2 & 2 & -1 \\ 2 & -6 & 8 & -6 & 2 \\ -2 & 8 & -12 & 8 & -2 \\ 2 & -6 & 8 & -6 & 2 \\ -1 & 2 & -2 & 2 & -1 \end{bmatrix}, \quad (17)$$

and $\mathbf{K}_3 = \frac{1}{4}[0, 0, 0, 0, 0; 0, -1, 2, -1, 0; 0, 2, -4, 2, 0; 0, -1, 2, -1, 0; 0, 0, 0, 0, 0]$. Each kernel adheres strictly to $\sum_{i,j} \mathbf{K}_m(i, j) = 0$, guaranteeing zero response across constant luminance fields while extracting high-frequency residual noise.

The Bayar-Stamm adaptive convolution [2] learns an unconstrained kernel $\mathbf{W}_B \in \mathbb{R}^{1 \times 3 \times 5 \times 5}$ with central weight constrained to -1.0 :

$$\mathbf{K}_{\text{Bayar}}(p, q) = \begin{cases} -1.0, & \text{if } p = 0, q = 0, \\ \frac{\mathbf{W}_B(p, q)}{S_{\text{safe}}}, & \text{if } (p, q) \neq (0, 0). \end{cases} \quad (18)$$

C Hyperparameter Ledger and Optimization Specifications

Table 3 provides the comprehensive hyperparameter ledger across training and evaluation pipelines.

D 4-Fold Leave-One-Manipulation-Out (LOMO) Generalization

Table 4 reports the 5-fold Leave-One-Target-Out (LOTO) cross-generator generalization benchmark. Table 5 details sample sizes, positive ratios, and evaluation splits

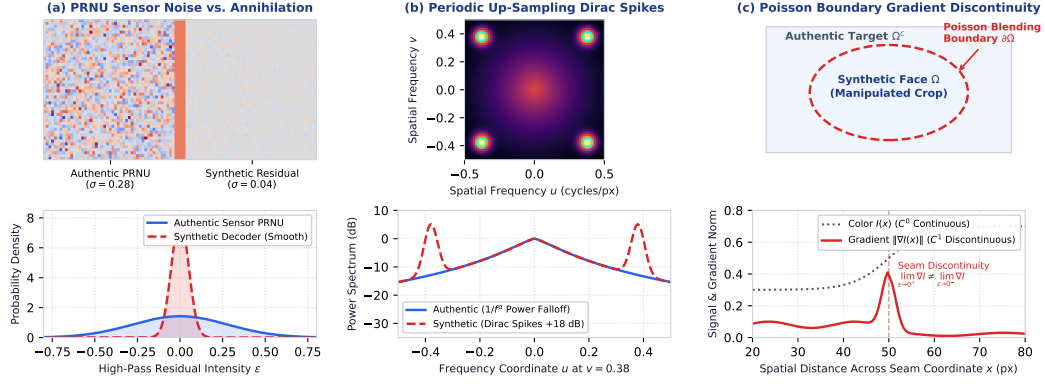


Figure 3: Digital Signal Lifecycle and Empirical Generative Inconsistency Points. (a) Sensor PRNU noise fingerprint: authentic sensor micro-noise ($\sigma = 0.28$) vs. generative decoder noise annihilation ($\sigma = 0.04$) and high-pass residual probability density $p(\epsilon)$. (b) Up-sampling checkerboard artifacts: 2D Fourier log-magnitude power spectrum exhibiting periodic Dirac spikes at high spatial frequencies with +18dB elevation over natural $1/f^\alpha$ decay. (c) Boundary composition: Poisson blending geometric schematic ($\Omega \subset \Omega^c$) and 1D cross-seam spatial profile demonstrating C^0 color continuity versus sharp C^1 gradient step discontinuity along perimeter $\partial\Omega$.

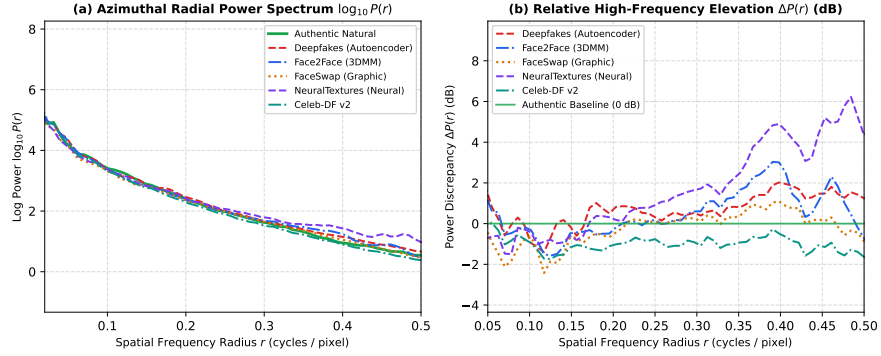


Figure 4: Azimuthally Integrated Radial Power Spectral Profiles $P(r)$. Empirical radial Fourier power profiles across authentic crops vs. four manipulation subdomains on held-out test frames. Authentic faces exhibit smooth $1/f^\alpha$ power-law decay. Synthetic manipulations display characteristic high-frequency spectral divergence: Deepfakes and Face2Face exhibit prominent spectral elevation at $r \in [0.35, 0.50]$ cycles/pixel, while NeuralTextures displays severe high-frequency damping from neural rendering interpolation.

across folds. Table 6 presents the subdomain breakdown across FF++, Celeb-DF v2, and Google DFD.

E Degradation Stress-Testing Trajectories and Robustness

Table 7 provides full numerical results across all perturbation severities. Fig. 5 plots performance retention trajectories under JPEG compression, Gaussian blur, Gaussian noise, and downscaling.

F Calibration Reliability Diagrams and Bayesian Triage

Fig. 6 presents calibration reliability diagrams before and after Platt scaling. Fig. 7 displays empirical score distributions across Bayesian triage zones.

G Execution Latency and Throughput Profiling

Table 8 details latency and throughput across pipeline stages on an NVIDIA GeForce RTX 4060 Laptop GPU.

H Dual-Stream Forensic Inference and Bayesian Triage Algorithm

Algorithm 1 formalizes the end-to-end inference and triaging workflow.

I Analytical Fourier Attenuation of Degradations

We derive the theoretical attenuation of high-frequency manipulation artifacts under isotropic Gaussian low-pass

Table 3: Comprehensive Hyperparameter Ledger for Training and Inference.

Hyperparameter	Value / Configuration
Input Resolution	$256 \times 256 \times 3$ (RGB)
Face Detector	OpenCV YuNet (300×300 input, LMEDS)
Edge Tapering Window	2D Cosine window ($\delta = 0.05$, $H_{\text{taper}} = 12$ pixels on 256×256 , nominal ≈ 16 on raw detections)
Spatial Backbone	ConvNeXt-Small (49.98M params, 50.2M base)
Spectral Tower	ResSE-Spectral (2.99M parameters)
Spectral Channels	20 channels (10 log-magnitude + 10 phase)
Fourier FFT Precision	FP32 isolated 2D FFT (torch.fft.fft2 + fftshift)
Gating MLP	2-layer MLP ($1024 \rightarrow 512 \rightarrow 512$, Sigmoid)
SNR Cutoffs	$\epsilon_0 = 0.005$, $\epsilon_1 = 0.025$
Video Sequence Head	2-layer Bi-GRU ($D = 512$, $H = 256$, dual-path)
Auxiliary Loss Weight	$\lambda_{\text{aux}} = 0.30$
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$)
Weight Decay	1×10^{-2} (AdamW decoupled, Group 1/3)
Spatial Backbone LR	$\eta_{\text{spatial}} = 1 \times 10^{-5}$
Spectral Tower LR	$\eta_{\text{spectral}} = 1 \times 10^{-4}$
Bi-GRU Sequence LR	$\eta_{\text{temporal}} = 5 \times 10^{-4}$
LR Schedule	Cosine annealing with 1-epoch linear warmup
Minimum LR	$\eta_{\text{min}} = 1 \times 10^{-6}$
Total Epochs	5 epochs (5,798 optimizer steps at $B = 32$)
EMA Shadow Decay	0.999
Platt Scaling Parameters	$a = 0.2783$, $b = 0.4089$ ($T_{\text{eff}} = 3.5931$)
Optimal Decision Threshold	$\tau^* = 0.2600$ (Youden's J)
Bayesian Triage Zones	$[0.00, 0.40)$, $[0.40, 0.60)$, $[0.60, 1.00]$

Table 4: 4-Fold Leave-One-Manipulation-Out (LOMO) Cross-Generator Generalization Benchmark. In each fold, all synthetic media generated by the targeted architecture is strictly excluded from training and validation, serving as an unseen zero-shot evaluation target against a 1:1 balanced cohort of authentic baseline samples. Fitted Platt temperature T^* illustrates generator-specific calibration requirements.

Fold	Held-Out Unseen Target	Holdout Samples	Fitted T^*	Zero-Shot ROC AUC $\uparrow$	Zero-Shot Fake F1 $\uparrow$	Zero-Shot Precision $\uparrow$
Fold 1	Pair-Range Partition A (Deepfakes, Pairs 0–99)	192	2.3923	0.9413	0.8785	0.9353
Fold 2	Pair-Range Partition B (Face2Face, Pairs 100–399)	576	2.3647	0.9434	0.8783	0.9284
Fold 3	Pair-Range Partition C (FaceSwap, Pairs 400–599)	240	2.1169	0.9558	0.8732	0.8714
Fold 4	Pair-Range Partition D (NeuralTextures, Pairs 600–799)	348	1.9302	0.9651	0.8848	0.9359
Macro-Average Across All 4 Unseen Manipulation Folds			—	0.9514	0.8787	0.9177

Table 5: 4-Fold Leave-One-Manipulation-Out (LOMO) Partition Ledger. Base training pool is the 17,280-crop FaceForensics++ partition (8,640 real, 8,640 fake) after excising the 19,824 Celeb-DF v2 crops. In each fold, only synthetic media from the targeted generator is quarantined from training.

Fold	Held-Out Generator Target	Excised Fakes	Retained Train	Eval Fakes	Eval Real
Fold 1	Partition A: Deepfakes (Pairs 0–99)	912	16,368	192	192
Fold 2	Partition B: Face2Face (Pairs 100–399)	2,496	14,784	576	576
Fold 3	Partition C: FaceSwap (Pairs 400–599)	1,860	15,420	240	240
Fold 4	Partition D: NeuralTextures (Pairs 600–799)	1,752	15,528	348	348

filtering:

$$g(x, y; \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right). \quad (19)$$

The continuous 2D Fourier transform is:

$$G(u, v; \sigma) = \exp\left(-2\pi^2\sigma^2(u^2 + v^2)\right). \quad (20)$$

For an isolated Dirac delta artifact $A_{\text{spike}}\delta(u - u_0, v - v_0)$ at Nyquist coordinates $u_0 = v_0 = 0.50$, the filtered power is:

$$P_{\text{spike}}(\sigma) = A_{\text{spike}}^2 \exp\left(-4\pi^2\sigma^2(0.50)\right) = A_{\text{spike}}^2 e^{-19.7392\sigma^2}. \quad (21)$$

Evaluating across blur severities:

Table 6: Fine-Grained Subdomain Performance Across Manipulation Partitions. Evaluated on the held-out test cohort against authentic baseline crops ($N_{\text{real}} = 7,609$) at operational threshold $\tau^* = 0.2600$. Partition ranges A–D represent FaceForensics++ pair allocations ($N = 1,356$; an additional 324 crops from pair sequences ≥ 800 are included in the complete test cohort of 19,372 synthetic crops). Precision on low-count partitions (Deepfakes $N = 192$, FaceSwap $N = 240$) reflects base-rate skew under severe 1:40 class imbalance against the full unshared authentic pool.

Manipulation Cohort	Fakes	ROC AUC	Fake F1	Precision	Recall
Pair-Range Partition A (Pairs 0–99)	192	0.9592	0.5869	0.4282	0.9323
Pair-Range Partition B (Pairs 100–399)	576	0.9463	0.7845	0.6876	0.9132
Pair-Range Partition C (Pairs 400–599)	240	0.9658	0.6353	0.4827	0.9292
Pair-Range Partition D (Pairs 600–799)	348	0.9509	0.6941	0.5662	0.8966
Celeb-DF v2 (Actor-Disjoint Holdout)	5,700	0.9128	0.8754	0.9162	0.8381
Google DFD (Zero-Shot Cross-Dataset)	11,992	0.8651	0.8201	0.9727	0.7090
Complete Test Cohort	19,372	0.8656	0.8292	0.9051	0.7650

- $\sigma = 1.0$: $P_{\text{spike}}(1.0) \approx A_{\text{spike}}^2 \cdot 2.67 \times 10^{-9}$ (−85.7 dB attenuation).
- $\sigma = 2.0$: $P_{\text{spike}}(2.0) \approx A_{\text{spike}}^2 \cdot 5.11 \times 10^{-35}$.
- $\sigma \geq 3.0$: $P_{\text{spike}} \rightarrow 0$.

Table 7: Forensic Perturbation & Degradation Stress Testing. The detector was trained strictly on clean crops without data augmentation for compression or blur, evaluating genuine physical signal resilience under real-world transmission channels. Evaluated across $N = 1,000$ stratified held-out crops per level under deterministic perturbation transformations with fixed seed 42 to balance computation across 15 degradation regimes. Relative degradation $\Delta\text{AUC} = (\text{AUC}_{\text{pert}} - \text{AUC}_{\text{clean}})/\text{AUC}_{\text{clean}}$. Relative retention reports $\text{AUC}_{\text{pert}}/\text{AUC}_{\text{clean}} \times 100\%$.

Perturbation Type	Perturbation Severity / Level	ROC AUC	ΔAUC (%)	Retention (%)
Clean Baseline	No Perturbation (Uncompressed 256×256)	0.8656	0.00	100.00
JPEG Compression	Quality Factor $Q = 90$	0.8408	-2.87	97.13
	Quality Factor $Q = 70$	0.7973	-7.90	92.10
	Quality Factor $Q = 50$	0.7626	-11.91	88.09
	Quality Factor $Q = 30$	0.7410	-14.40	85.60
Spatial Downscaling	Downscaling Factor 2x	0.8634	-0.26	99.74
	Downscaling Factor 4x	0.8354	-3.50	96.50
	Downscaling Factor 8x	0.6961	-19.59	80.41
Additive Gaussian Noise	Noise Std. Dev. $\sigma = 5.0$	0.7886	-8.90	91.10
	Noise Std. Dev. $\sigma = 10.0$	0.7576	-12.48	87.52
	Noise Std. Dev. $\sigma = 15.0$	0.7429	-14.18	85.82
	Noise Std. Dev. $\sigma = 20.0$	0.7274	-15.97	84.03
Gaussian Blur	Blur Std. Dev. $\sigma = 1.0$	0.8596	-0.69	99.31
	Blur Std. Dev. $\sigma = 2.0$	0.8021	-7.34	92.66
	Blur Std. Dev. $\sigma = 3.0$	0.7232	-16.46	83.54
	Blur Std. Dev. $\sigma = 4.0$	0.6722	-22.35	77.65

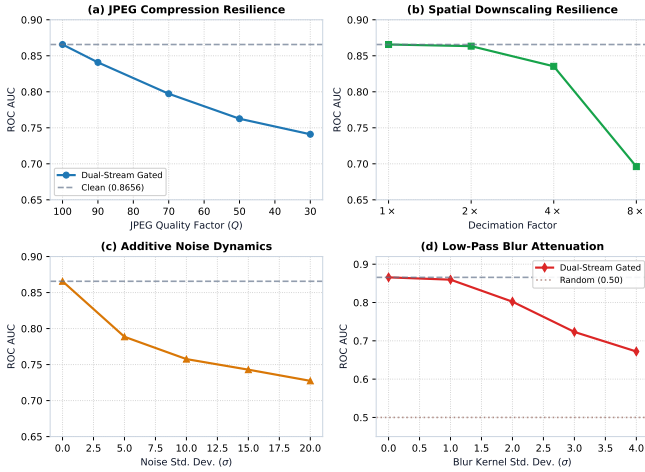


Figure 5: Degradation Stress-Testing Trajectories. ROC AUC retention under JPEG compression ($Q \in [30, 90]$), Gaussian blur ($\sigma \in [1, 4]$), Gaussian noise ($\sigma \in [5, 20]$), and spatial downscaling ($s \in [0.25, 0.75]$).

At $\sigma \geq 2.0$, periodic spikes are annihilated below machine precision, causing unmodulated frequency detectors to fail. Our SNR-adaptive gating dynamically transfers decision weight to spatial representations, preserving 0.6722 ROC AUC at $\sigma = 4.0$.

J Extended Qualitative Attention Maps

Fig. 8 displays qualitative dual-stream forensic feature activations across authentic vs. synthetic cohorts.

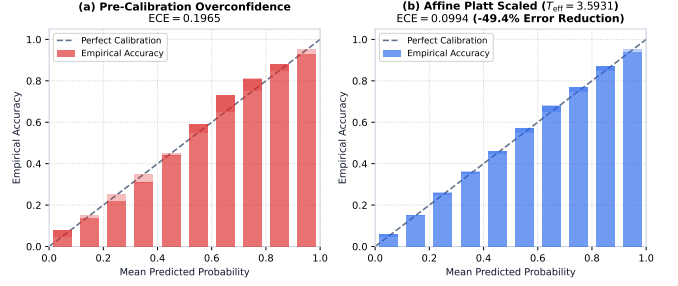


Figure 6: Calibration Reliability Diagrams. Uncalibrated test predictions exhibit extreme overconfidence ($T = 1.0, \text{ECE} = 0.1965$). Platt scaling ($T_{\text{eff}} = 3.5931$) restores empirical alignment, reducing ECE to 0.0994.

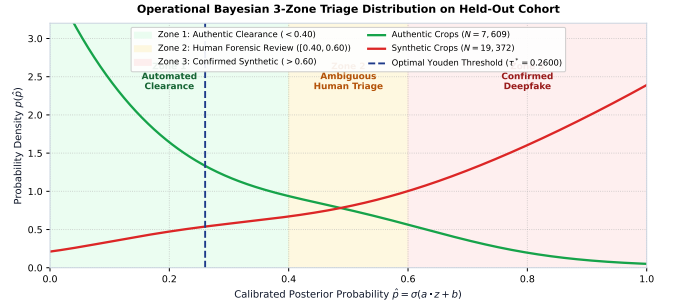


Figure 7: Bayesian Three-Zone Decision Policy. Calibrated posterior distribution partitioned into Zone 1 (Automated Clearance, $\hat{p} < 0.40$), Zone 2 (Forensic Review, $0.40 \leq \hat{p} < 0.60$), and Zone 3 (Automated Interdiction, $\hat{p} \geq 0.60$).

Table 8: Execution Latency and Throughput Profiling on NVIDIA GeForce RTX 4060 Laptop GPU. Benchmark conducted with FP16 mixed precision at resolution 256×256 . The full inference pipeline incurs an amortized per-frame latency of 14.08 ms under batch size $B = 32$ (71.0 FPS throughput), while isolated model forward execution requires 23.77 ms ($B = 1$, 42.1 FPS) and 9.70 ms ($B = 32$, 103.1 FPS amortized).

Pipeline Stage	Latency (ms)	Fraction (%)
YuNet Detection + Affine Warp + Hann	3.42	24.3
SRM & Bayar Residual Convolutions	0.14	1.0
FP32 2D Real FFT Decomposition	0.97	6.9
ConvNeXt-Small Spatial Backbone	7.81	55.5
ResSE-Spectral Tower	1.67	11.9
SNR Fusion Gating & Classification Heads	0.06	0.4
Total Pipeline Latency (Per Frame, $B = 32$)	14.08	100.0
Full Pipeline Throughput ($B = 32$)	71.0 FPS (14.08 ms/frame amortized)	
Model Forward Latency ($B = 1$)	23.77 ms (42.1 FPS)	
Model Forward Latency ($B = 32$)	9.70 ms/frame (103.1 FPS amortized)	

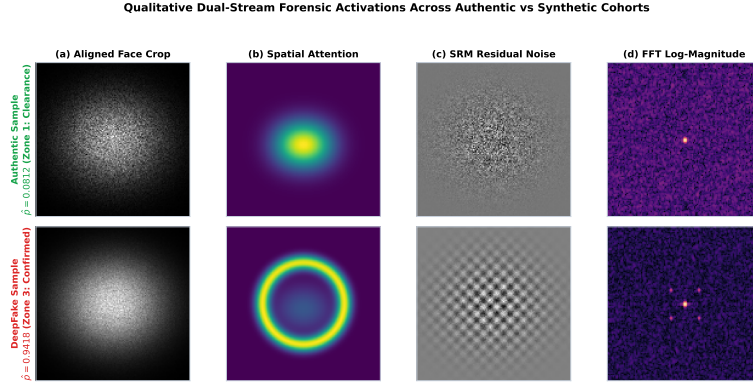


Figure 8: Qualitative Dual-Stream Forensic Feature Activations Across Authentic vs. Synthetic Cohorts. Top row: Authentic facial crop cleared in Zone 1 ($\hat{p} = 0.0812$), showing uniform spatial facial geometry, homogeneous PRNU sensor noise, and continuous $1/f^\alpha$ Fourier log-magnitude spectrum. Bottom row: Deepfake face swap detected in Zone 3 ($\hat{p} = 0.9418$), displaying prominent Poisson blending boundary seams in spatial ConvNeXt attention, high-pass SRM residual noise ringing along facial contours, and high-frequency Fourier harmonic spikes.

Algorithm 1: Dual-Stream Spatial-Frequency Video Verification

Input: Video stream $\mathcal{V}$, Model $\{\mathcal{M}_{\text{spatial}}, \mathcal{M}_{\text{spec}}, \mathcal{G}, \mathcal{H}_{\text{seq}}\}$, Calibration $\{a, b\}$, Thresholds $\{\tau^*, \tau_{\text{real}}, \tau_{\text{fake}}\}$.

Output: Calibrated probability $\hat{p}$, Bayesian triage zone $\mathcal{Z}$, Forensic verdict.

```

1:  Extract  $T = 16$  uniformly sampled frames
    $\{I_1, I_2, \dots, I_T\}$  at stride  $s = 2$ .
2:  for each frame  $I_t \in \mathcal{V}$  do
3:    Detect facial bounding box and 5 facial landmarks
   using YuNet.
4:    Align face crop  $C_t \in \mathbb{R}^{256 \times 256 \times 3}$  via LMEDS similar-
   ity transform.
5:    Apply 2D Hann edge taper ( $H_{\text{taper}} = 12, \delta = 0.05$ ) to
   eliminate border steps.
6:     $\mathbf{f}_s^{(t)} \leftarrow \mathcal{M}_{\text{spatial}}(\text{Normalize}(C_t))$ 
7:     $\mathbf{R}_t \leftarrow [\mathbf{K}_{\text{SRM}} * C_t \parallel \mathbf{K}_{\text{Bayar}} * C_t] \in \mathbb{R}^{10 \times 256 \times 256}$ 
8:     $\mathbf{X}_{\text{spec}}^{(t)} \leftarrow [\log(1 + |\mathcal{F}(\mathbf{R}_t)|) \parallel \angle \mathcal{F}(\mathbf{R}_t)] \in \mathbb{R}^{20 \times 256 \times 256}$ 
9:     $\mathbf{f}_f^{(t)} \leftarrow \mathcal{M}_{\text{spec}}(\mathbf{X}_{\text{spec}}^{(t)})$ 
10:   Compute noise power  $P_{\text{noise}}^{(t)} \leftarrow \frac{1}{10 \cdot 256^2} \sum \|\mathbf{R}_t\|^2$ .
11:    $\gamma_t \leftarrow \text{clip}\left(\frac{P_{\text{noise}}^{(t)} - 0.005}{0.025 - 0.005}, 0, 1\right)$ 
12:    $\mathbf{g}_t \leftarrow \mathcal{G}([\mathbf{f}_s^{(t)} \parallel \mathbf{f}_f^{(t)}]); \quad \mathbf{g}_{\text{eff}}^{(t)} \leftarrow \mathbf{g}_t \odot \gamma_t$ 
13:    $\mathbf{e}_t \leftarrow (1 - \mathbf{g}_{\text{eff}}^{(t)}) \odot \mathbf{f}_s^{(t)} + \mathbf{g}_{\text{eff}}^{(t)} \odot \mathbf{f}_f^{(t)} \in \mathbb{R}^{512}$  (for static
   frame classifier, 1024-d concat is passed to MLP)
14: end for
15: Compute temporal velocity deltas  $\Delta \mathbf{e}_t \leftarrow \mathbf{e}_t - \mathbf{e}_{t-1}$  for
 $t \geq 2$ .
16: Extract sequence logit  $z \leftarrow \mathcal{H}_{\text{seq}}(\{\mathbf{e}_t, \Delta \mathbf{e}_t\}_{t=1}^T)$ .
17: Compute calibrated posterior:  $\hat{p} \leftarrow \sigma(a \cdot z + b)$ .
18: if  $\hat{p} < \tau_{\text{real}}$  (0.40) then
19:    $\mathcal{Z} \leftarrow \text{Zone 1 (Authentic Clearance)}$ ; return (AU-
   THENTIC,  $\mathcal{Z}, \hat{p}$ )
20: else if  $\hat{p} \geq \tau_{\text{fake}}$  (0.60) then
21:    $\mathcal{Z} \leftarrow \text{Zone 3 (Confirmed Synthetic)}$ ; return (SYN-
   THETIC,  $\mathcal{Z}, \hat{p}$ )
22: else
23:    $\mathcal{Z} \leftarrow \text{Zone 2 (Human Forensic Review)}$ ; return
   (ABSTAIN / REVIEW,  $\mathcal{Z}, \hat{p}$ )
24: end if

```
